INCH-POUND

MIL-DTL-32723
09 March 2023

DETAIL SPECIFICATION

MULTI-PURPOSE, PERSONAL, HYDRATION SYSTEM

This document is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers a Chemical, Biological, Radiological, Nuclear (CBRN) resistant, Multi-Purpose Personal Hydration System (MPHS), and a MPHS training kit to simulate use of the CBRN hydration system in a training environment with a non CBRN resistant hydration system.

1.2 Classification. The multi-purpose personal hydration systems covered under this document are of the following types:

1.2.1 Types.

Type I - CBRN Hydration System
Type II - CBRN Hydration Training Kit
(Includes one (1) of each Type IIA, Type IIB and Type IIC)
Type IIA - Bite valve with cover
Type IIB - Mask adapter
Type IIC - Drink tube adapter

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2. APPLICABLE DOCUMENT

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-32380 - Bags-Unit Pack Barrier, Chemical, Biological Protective Clothing and Accessories

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY, DEVCOM - SOLDIER CENTER

2-6-1023 - Bladder, MPHS Hydration System, Chemical/ Biological

2-6-1030 - HL Hydrolock, Gen 2, Lot Control, Chemical/ Biological Hydration Unit

2-6-1040 - HL Type A GMA, Lot Control, Chemical/ Biological Hydration Unit

2-6-1042 - Bite Valve and Cover Assembly Chemical/ Biological Hydration Unit

2-6-1043 - Training Kit four (4)-PC

(Copies of drawings are available from the U.S. Army, DEVCOM Soldier Center, ATTN: FCDD-SCP-PEI, 10 General Greene Avenue, Natick, MA 01760-5019.)

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OTHER GOVERNMENT DOCUMENTS AND PUBLICATIONS

PRF EA-M-10003A, System Specification for Mask, Joint Service General Purpose (JSGPM)

(Copies of this document are available through the contracting activity.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles

(Copies of this document are available online at https://aatcc.org.)

ASTM INTERNATIONAL

ASTM F88/F88M - Standard Test Method for Seal Strength of Flexible Barrier Materials
ASTM D882 - Standard Test Method for Tensile Properties of Thin Plastic Sheeting
ASTM D3218 - Standard Specification for Polyolefin Monofilaments

(Copies of these documents are available online at https://www.astm.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://www.asq.org.)

SAE INTERNATIONAL

SAE-AMS-STD-595 – Colors used in Government Procurements

(Copies of this document are available online at https://standards.sae.org/amsstd595/).

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Unless a specific exemption has been obtained, nothing in this document, supersedes applicable laws and regulations.

3. REQUIREMENTS

3.1 Validation. Prior to contract award, MPHS items shall be validated through test and inspection (Section 4) by the Program Management Office (PMO). A list of all approved Government Laboratories will be available from the PMO (see 6.6).

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3.1.1 First article testing (FAT). When specified (see 6.2), a sample shall be subjected to first article inspection and testing in accordance with 4.2. As specified in the contract, all initial purchases shall require a FAT, or a FAT waiver shall be submitted to the Government for approval or rejection. During FAT, the Government shall approve the color samples and provide approved standard samples with letter of acceptance to the contractor to use as the standard to match for follow on production color matching (see 3.5).

3.1.1.1 Chemical resistance performance. First Article Chemical Testing (FACT) and Production Lot Chemical Testing (PLCT) shall be conducted on full assembled end items. Failure to meet the chemical requirements or any sub-test will constitute failure for the entire FAT or Production Lot Testing (PLT).

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 End item requirements. The MPHS materials, design, and performance shall conform to and pass the applicable specifications, standards, drawings and/or patterns, and testing required herein as stated in Table I.

TABLE I. Testing and inspection requirements.

Characteristic	Requirement	FAT 1/	CI/ PLT 1/	Verification Method	Government Conformance Inspection* 1/
Type I - CBRN Hydration System					
End item components	3.4.1	X	X	Drawing 2-6-1023	COC
Materials	3.4.1.1	X	X	Drawing 2-6-1023	COC
Color	3.4.1.1	X	X	4.5.1	IPV/COC
Tensile strength	3.4.1.2	X	X	4.5.2	COC
Seam strength	3.4.1.3	X	X	4.5.3	COC
Chemical resistance	3.4.1.4	X	X	4.5.4	COC
Resistance to fungus	3.4.1.5	X		4.5.5	COC
Breakaway resistance	3.4.1.6	X	X	4.5.6	IPV/COC
Freezing	3.4.1.7	X		4.5.7	COC
Empty weight	3.4.1.8	X	X	4.5.8	IPV
Functional integration	3.4.1.9	X	X	4.5.9	IPV

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TABLE I. Testing and inspection requirements. – Continued

Characteristic	Requirement	FAT 1/	CI/ PLT 1/	Verification Method	Government Conformance Inspection* 1/
Type I - CBRN Hydration System – Continued					
Leakage	3.4.1.10	X	X	4.5.10	IPV
Capacity	3.4.1.11	X	X	4.5.11	IPV
Compatibility with current mask	3.4.1.12	X	X	4.5.12	IPV
Type II - CBRN Hydration Training Kit					
End item components	3.4.2	X	X	Drawing 2-6-1043	COC
Materials (Type II)	3.4.2.1	X	X	Drawing 2-6-1043	COC
Type IIA - bite valve with cover	3.4.2.2	X	X	Drawing 2-6-1042	COC
Type IIB - Mask adapter	3.4.2.3	2B X	3B X	Drawing 2-6-1040	COC
Type IIC - drink tube adapter	3.4.2.4	4BX	5BX	Drawing 2-6-1030	COC
Type I - CBRN Hydration System and Type II - CBRN Hydration Training Kit					
Manufacturing and assembly	3.6.1 & 3.6.2	X	X	Drawings 2-6-1023 (Type I) 2-6-1043 (Type II)	IPV
Identification markings and labeling	3.7.1	X	X	4.5.13 and Table II	IPV
User instructions	3.7.2	X	X	4.5.14 and Table II	IPV
Barrier bags	3.8	X	X	4.5.15	IPV/COC
1/ LEGEND:					
X - Requires FAT or PLT from Government testing facility or laboratory or Conformance Inspection (CI) at contractor's facility by Government Quality Assurance Representative (QAR).					
COC - Certificate of Conformance documentation for required testing not performed at Government test facility. When COCs are required, testing data documentation must accompany the COC. The Government reserves the right to inspect such items to determine the validity of the certification.					
IPV - In-process validation. Contractor to perform test and inspections throughout production in accordance with the protocols established in their Quality Management System (see 4.3.1 and 4.5.16).					
* - Government QAR ONLY will perform CI at contractor's facility on samples selected for Chemical Lot Acceptance Testing as well as sampling performed for IPV.					

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3.3.1 Production lot definition. A MPHS end item lot shall consist of homogeneous chemical barrier layer embedded within the bladder laminate film. Only whole (complete and unused) lots of chemical barrier layer shall be used within the laminated film. Multiple lots of barrier laminate film may be used within a MPHS production lot, not to exceed the representative quantities per each homogeneous barrier lot, required for sample selection for chemical testing. Each production lot shall contain chemical barrier lots specific to itself. A new lot shall be established following the occurrence of any event affecting the homogeneity of the lot.

3.4 Components and performance.

3.4.1 Type I - CBRN hydration system. All components for Type I shall conform to the specifications and drawings listed on Drawing 2-6-1023 (see 2.2.2). This system shall contain a hydration reservoir (or bladder) with drink tube, bite valve with cover, JSGP Mask drink tube adapter as specified in PRF EA-M-10003A, instructions, and be packaged in CBRN resistant packaging. Due to the CBRN protection requirement, MPHS contains critical components whose failure during operational use could cause loss of life, permanent disability, or major injury and therefore is designated a Critical Safety Item (CSI). All Government administrative and surveillance procedures applicable to CSI items shall be invoked in accordance with the contract or purchase order.

3.4.1.1 Type I material requirements. All materials and color for Type I cited on Drawing 2-6-1023 shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings and shall meet the requirements of 3.4.1.2 to 3.4.1.12 when tested as specified in 4.5.

3.4.1.2 Tensile strength bladder material. The tensile strength of the Type I bladder material shall be a minimum of 50-pounds per square inch and shall have a minimum elongation of 400 percent when tested as specified in 4.5.2.

3.4.1.3 Seam strength of bladder. The seam strength of the seams of the Type I bladder shall be a minimum of 25-pounds per square inch and a minimum elongation of 150 percent when tested as specified in 4.5.3.

3.4.1.4 Chemical resistance. The Type I shall prevent contamination of contents by Distilled Mustard (HD) and Thickened Soman (TGD), Chemical Warfare Agents when tested as specified in 4.5.4.

3.4.1.5 Resistance to fungus. The Type I shall be resistant to fungus growth when tested as specified in 4.5.5.

3.4.1.6 Breakaway resistance. The Type I shall resist breakaway at various interfaces when tested as specified in 4.5.6.

3.4.1.7 Freezing. The Type I shall remain intact and not leak when tested as specified in 4.5.7.

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3.4.1.8 Empty weight. The Type I when empty shall weigh no more than 0.99-pounds when tested as specified in 4.5.8.

3.4.1.9 Functional integration. The Type I shall be sized to fit within the 63-millimeter opening of the Modular Lightweight Load-Carrying Equipment (MOLLE) Hydration System Carrier (HSC) as specified in CO-PD-02-02 and the Aviation Hydration Carrier (AHC) as specified in GL-PD-19-04 when tested as specified in 4.5.9.

3.4.1.10 Leakage. The Type I when filled to capacity and compacted at 500-pound force shall show no evidence of leakage when tested as specified in 4.5.10.

3.4.1.11 Capacity. The Type I shall have a capacity of 3-liters (± 133-milliliters) or 101.442 (± 4.5) ounces when tested as specified in 4.5.11.

3.4.1.12 Compatibility with masks (Type I only). The Type I shall connect to the current chemical protection mask drinking system and shall meet a minimum flow rate of 200-milliliters per minute at 3-pounds per square inch when tested as specified in 4.5.12.

3.4.2 Type II - CBRN hydration training kit. The Type II training kit includes one (1) of each Type IIA, Type IIB and Type IIC. This system shall contain the necessary components for connecting to the JSGP Mask as specified in PRF EA-M-10003A and to non-CBRN hydration systems for use by Soldiers in a training environment only. The Type II CBRN hydration training kit shall contain three (3) items, one (1) Type IIA Bite valve with cover, one (1) Type IIB – Mask adapter, and one (1) Type IIC – Drink tube adapter. All components for the Type II shall conform to the specifications and drawings listed on Drawing 2-6-1043 (see 2.2.2).

3.4.2.1 Type II material requirements. All materials and color for Type II cited on Drawing 2-6-1043 shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings. Note: This system does not provide CBRN protection and is for use by Soldiers in a training environment only.

3.4.2.2 Type IIA - Bite valve with cover. The bite valve with cover which is a component of the CBRN hydration training kit, shall conform to Drawing 2-6-1042. This item allows the user to drink water through the bite valve. The bite valve with cover is used by Soldiers in a training environment only.

3.4.2.3 Type IIB - Mask adapter. The mask adapter which is a component in the CBRN hydration training kit, shall conform to Drawing 2-6-1040. This item connects to the drink tube adapter and JSGP Mask as specified in PRF EA-M-10003A. The mask adapter is used by Soldiers in a training environment only.

3.4.2.4 Type IIC - Drink tube adapter. The drink tube adapter is a component of the CBRN hydration training kit and shall conform to Drawing 2-6-1030. This item connects to the MOLLE Hydration System and the Aviation Hydration System and can also connects to Types IIA and IIB. The drink tube adapter is used by Soldiers in a training environment only.

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3.5 Production lot color testing. The color and appearance shall be as specified in the contract and shall match the Government approved standard sample(s) during first article when tested as specified in 4.5.1.

3.6 Manufacture and assembly.

3.6.1 Type I. The Type I MPHS shall be manufactured and assembled as shown on Drawing 2-6-1023 (see 4.3.1).

3.6.2 Type II. The Type II MPHS shall be manufactured and assembled as shown on Drawing 2-6-1043 (see 4.3.1).

3.7 Identification markings, labels and user and care instructions.

3.7.1 Identification markings and labeling (Type I only). The markings on the face side of the Type I bladder shall be permanently and legibly marked in letters of not less than 1/4-inch and not more than 1/2-inch in height with either yellow or white indelible marking ink/solution. The use of adhesive labels is prohibited. The bladder shall be labeled in accordance with; acquisition documents, Drawing 2-6-1023 and subsidiary drawings. Identification markings shall be inspected as specified in 4.5.13.

3.7.2 User instructions (Type I and Type II). User instructions will be provided with the acquisition documents. Instructions shall be easily readable, undamaged, and included in MPHS packaging per 4.5.14.

3.7.3 Care instructions. Scrape dirt and dust from the item using a brush that will not cut the fabric or other material. Wash the items with mild detergent or soap and rinse with clean water. Do not use chlorine bleach, yellow soap, cleaning fluid, or solvents that will discolor or deteriorate the item. Do not launder or dry item in fixed field, commercial or home-type laundry equipment. Do not attempt to dye or repair. Turn in any damaged item for replacement.

3.8 Unit pack barrier bags (Type I). The Type I shall be individually packaged in unit pack barrier bags. The unit pack barrier bag shall be 8.5 (± 1) inches wide and 20 (± 1.5) inches tall. The unit pack barrier bag shall be constructed and tested in accordance with Type II of MIL-PRF-32380 with the following exceptions (see 4.5.15).

1. The bags shall be made from either flexible, multi-layer formable foil film with an opaque finish, or from flexible, multi-layer nylon foil film with an opaque finish. They shall have a tear, nick, or “V” notch, 1/8-inch deep in at least one (1) edge, 2-inches from the end of the bag. The bags shall have the properties listed below:

   a. The formable foil film shall be constructed as follows:
      0.0012-inch Oriented Polypropylene Adhesive Layer
      0.00175-inch Aluminum Foil Adhesive Layer
      0.003-inch Cast Polypropylene

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b. The nylon foil film shall be constructed as follows:
60g Nylon
0.0005-inch Polyethylene
0.00035-inch Aluminum Foil
0.0022-inch Polyethylene

2. The foil film color shall be Gray 16350 in accordance with SAE-AMS-STD-595.
3. The tensile strength shall be a minimum of 24-pounds in the machine direction and 20-pounds in the cross-machine direction.
4. The tear resistance shall be a minimum of 140-grams in the machine direction and 100-grams in the cross-machine direction.
5. The seal strength shall be a minimum of 10-pounds per inch.
6. The ultraviolet resistance test shall be after 250 hours of exposure in accordance with ASTM D3218.

3.8.1 Unit pack barrier bag markings. The markings shall be applied to the unit pack barrier bag. Adhesive labels shall not cause any deleterious effect to the barrier bag material and shall provide 100 percent adhesion for the lifetime of the product. The height of lettering shall be 1/4 to 1/2-inch. The unit pack barrier bag markings shall be located on the face side of the unit pack barrier bag and shall include the following information:

NATIONAL STOCK NUMBER: 8465-01-643-5606 CBRN HYDRATION SYSTEM, MPHS CONTRACT NO: DATE PACKAGED: *MM/YY* MANUFACTURE DATE: *MM/YY* SURVEILLANCE INSPECTION DATE: *MM/YY* 1/ SURVEILLANCE MARKING NO: *SLBYYMXXXXXX* 2/ CAUTION: KEEP UNOPENED BAG INSIDE CLOSED SHIPPING CONTAINER. DO NOT EXPOSE BAG TO PROLONGED SUNLIGHT.

1/ Surveillance Inspection Date shall be five (5) years from Manufacture Date.
2/ The surveillance marking number shall be 12 characters consisting of the following:

Location of Manufacture:
SLB – Three alpha characters

Manufacture date:
YYM the year of production as shown as a numeric code, consisting of the last two (2) digits of the calendar year in which the manufacture, assembly, or regrouping of the lot was initiated. The Month of production is an alpha code to identify the month that manufacture, assembly, or regrouping of the lot was initiated. The single alpha code assigned for each month shall be as follows:

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January	- A	July	- G
February	- B	August	- H
March	- C	September	- J
April	- D	October	- K
May	- E	November	- L
June	- F	December	- M

3.8.2 Barrier bag (Type II). Each Type II training kit shall be packaged in clear, resealable press seal plastic bag or as specified in the contract or purchase order.

3.9 Workmanship. The finished MPHS shall be uniform in quality and shall conform to the quality of product established by this specification. The occurrence of defects, as specified in Table II, shall not exceed the criteria specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).
c. Production lot testing (4.4).

4.2 First article inspection. A first article, submitted in accordance with 3.1.1 shall be examined for the defects specified in Table II, the verification tests for the characteristics as specified in Table I, and the tests specified in 4.5.1 through 4.5.15.

4.2.1 Chemical resistance re-validation testing. At any point after a first article inspection has been approved (see 3.1.1), any material or process change to the approved MPHS design must be tested in accordance with 4.5.4 and approved by the Government. Samples shall be tested in a Government approved lab. A list of all approved Government laboratories will be available from the PMO (see 6.6).

4.3 Conformance inspection (CI). Conformance inspection shall include the visual examination of defects in Table II, the tests in Table I and tests in 4.5.1 through 4.5.15 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 with acceptance quality limits (AQLs) as specified in the contract or order, except where otherwise indicated (see 6.2).

4.3.1 Certificate of conformance (COC). When COC(s) is required, testing data documentation must accompany the COC. The Government reserves the right to inspect such items to determine the validity of the certification.

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4.3.2 Inspection conditions. Unless otherwise specified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.4 End item production lot testing (PLT). The Government will perform CI on end items samples selected to be sent for chemical testing. Contractor shall show evidence that all contiguous lots of barrier film are represented in the sample, proportional to their distribution in production lot. COC shall be presented to the Government at the time of end item conformance inspection. Results of both Chemical testing, inspections, and examinations will determine the acceptance or rejection of entire production lot. Additionally, when a lot fails PLT and is subsequently rejected, that lot is rejected in total, and no component parts may be used in the production of any other lot. Failed lots shall not be delivered and shall not be co-mingled with other lots. Government shall require documented evidence that rejected items will not get back into the DoD supply system, General Services Administration (GSA), and other venues in which Soldiers may buy replacements using individual or unit funds. The Government shall have the right to revoke acceptance of any and all items that may contain the root cause failure mode, and to require the Contractor to replace all affected units at the Contractor's expense, including transportation costs. Failures occurring during Government testing are also subject to this requirement.

4.4.1 End item inspection. At the conclusion of a PLT, all items shall be available for CI and follow-on chemical testing (see appendix A). The Contractor shall perform end item inspection per the approved Quality Management Plan (QMP), and this document.

4.4.2 End item visual examination. The end item shall be examined for the defects listed in Table II. The sample unit shall be one (1) MPHS. The finding of one (1) or more critical defects shall be cause for failure of the lot.

4.4.2.1 End item critical visual defect examination. Prior to performing the end item visual examination in accordance with 4.4.2, 100 percent of the end items shall be examined for the critical defects listed in Table II. This 100 percent examination for critical defects should be a part of the contractor's inspection system or quality assurance program. Any item found to contain one (1) or more critical defects, shall be a failure of the entire lot.

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TABLE II. Visual examination of defects.

Examine	Defect	Classification		
		Critical	Major	Minor
Construction and workmanship - Type I only				
Bladder Assembly	Any cut, tear, hole, rip, crack, or rupture through the bladder material	01		
	Any visible void	02		
	Radio Frequency weld not complete or missing in areas	03		
	Any burn spot which cracks on flexing, bending, or stretching (by hand)	04		
	Any pinch, pit, thin spot, abraded area, deep crease, readily removable foreign matter, or similar defective condition which results in a thickness less than the minimum thickness allowed for bladder	05		
	Any foreign matter within weld area or bladder material	06		
	Any repair or patch	07		
	Any solid ridge, run or lump resulting in a thickness greater than the maximum thickness of the film allowed for the bladder		101	
	Any malformation or distortion		102	
	Any component part omitted or not as specified		103	
	Any components failing to serve their intended purpose	08		
Tubing	Any hole, cut, cracking, or deformed	09		
	Any components failing to serve their intended purpose	10		
Cap assembly	Stripped threading, cracking, damaged, deformed, or hole		104	
	Fill cap lanyard loose		105	
	Any visibly missing, cracked, or loose O-rings		106	
Unit pack barrier bag	Per MIL-PRF-32380 NOTE: All defects for unit pack barrier bags will be measured as a Major or Minor defect and will not automatically fail a MPHS production lot.		--	--
Marking and labeling	Missing, incomplete, illegible, or incorrect on any components except for molded components (see above)			201
	Not indelible (permanent)		107	

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TABLE II. Visual examination of defects. – Continued

Examine	Defect	Classification		
		Critical	Major	Minor
Construction and workmanship - Type I only - Continued				
Functional integration	End item assembly not interchangeable with the MOLLE HSC and the AHC (see 4.5.9)	11		
Construction and workmanship – Type II only				
Barrier bag	Bag ripped, torn, or cannot be zipped closed			202
Construction and workmanship – Type I and Type II				
Adapters	Any hole, chip, broken, or cracking	12		
	Any malformation of molded pieces	13		
	Excessive molding debris			203
	Any components failing to serve their intended purpose	14		
	Any visibly loose O-rings		108	
	Molded numbers, letters, or symbols not readable			204
Care Instructions	Omitted, incorrect, illegible		109	
Color	Not matching color of approved standard samples or drawings	15		
Compatibility	End item assembly attachments not interchange-able for use with the current mask assemblies (see 4.5.12)	16		
Dimensions (overall)				
Bladder (Type I only)	Does not meet critical dimensions within specified tolerances on drawings	17		
Tubing (Type I only)	Does not meet critical dimensions within specified tolerances on drawings	18		
Components (Type I & Type II)	Does not meet critical dimensions within specified tolerances on drawings		110	

4.4.2.2 Bladder components. Bladder components will be 100 percent inspected by Contractor during the cutting process to determine that parts containing defects (see Table II) are removed from production. Hoses and plastic molded attachment/components shall be 100 percent inspected by contractor during the assembly of the individual items to ensure they do not contain any defect noted in Table II. Repairs are not authorized on any chemical resistance materials or components. Rework or use-as-is may be authorized on noncritical components or subassemblies, but Contractor must submit nonconformance to the Government for disposition approval.

4.5 End item verification testing. The items shall be tested for the characteristics shown in Table I except for Chemical Resistance, which will be covered by paragraph 3.4.1.4 and 4.5.4. The conformance inspection and verification testing lot size shall be expressed in units of one

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MPHS. The sample unit shall be one MPHS unit. All requirements as noted in Table I shall be applicable to the sample unit. All test reports shall contain the individual values used in expressing the final result. The methods of testing as specified wherever applicable and as listed in Table I shall be followed. Except for the first article testing, the sample size shall be as specified below. One (1) or more sample units failing to meet any specified requirement shall be cause for rejection of the lot.

Lot size (MPHS)	Sample size (each)
Up to 6,000	15
6001 to 12,000	22
12,001 to 20,000 (max.)	33

4.5.1 Production lot color testing. The color and appearance of the MHPS shall match the standard sample when viewed using the AATCC EP 9, Option C, (see 6.6) with a primary light source simulating the spectral quality of average daylight, CIE Illuminant D65, with a color temperature of 6500 (± 200) Kelvin (K) illumination of 100 (± 20) foot candles. Alternatively, the color and appearance of the MHPS shall match the standard sample with a primary light source simulating artificial daylight, CIE Illuminant D75, with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles in lieu of D65. (See 6.6.1). The MHPS shall also be a good match to the standard sample with a secondary source simulating the spectral quality of incandescent lamplight, CIE Illuminant A, with a color temperature of 2856K (± 200).

4.5.2 Tensile strength (Type I only). Tensile strength for the bladder shall be measured in accordance with ASTM D882. A minimum of five (5) samples shall be tested from each bladder. The individual results for each sample shall be reported.

4.5.3 Seam strength (Type I only). Seam strength for the bladder shall be measured in accordance with ASTM F88/F88M. A minimum of five (5) samples shall be tested from each bladder. The individual results for each sample shall be reported.

4.5.4 Chemical resistance (Type I only). The chemical resistance testing shall be conducted by a Government approved source as specified in the contract. System level chemical agent testing shall be tested in accordance with Appendix A. A total of 14 MPHS systems shall be tested. Seven (7) MPHS systems shall be tested against HD with the currently issued mask adapter installed. Seven (7) MPHS systems shall be tested against TGD with the currently issued mask adapter installed. If upon failure analysis review, the Government approves retesting, the addition of a carrier will be authorized if sought by Contractor.

4.5.5 Resistance to fungus (Type I only). Three (3) bladders shall be tested in accordance with MIL-STD-810, Method 508.6 for resistance to fungus. The following five (5) species shall be used for this test: Aspergillus niger, Aspergillus flavus, Aspergillus versicolor, Penicillium funiculosum, and Chaetomium globosum. A cotton fabric control item shall be used to verify appropriate spore-growing conditions exist. One (1) of the three (3) test items shall be cut to expose the interior of the bladder. The control item and test items shall be inoculated with the mixed fungus spore suspension by spraying the suspension on the items in the form of a fine mist from an atomizer or nebulizer. The test period shall be 28 days. After seven (7) days from the start of the test, the

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control item (cotton fabric) shall be inspected to ensure conditions were suitable for microbial growth. There shall be no more than light fungus growth when visibly examined.

4.5.6 Breakaway resistance (Type I only). The tubing shall be subjected to a pull test from the opening on the bladder. The bladder shall be held in the lower jaws of an Instron (or similar) test machine. The tubing shall be held in the upper jaws. The initial separation between the jaws shall be 4-inches (gauge length). The jaw separation speed shall be 12-inches per minute. The test will be conducted by pulling the tubing from the opening. Three (3) samples shall be tested. All three (3) samples must have a minimum separation force of 50 pounds.

4.5.7 Freezing (Type I only). The bladder shall be filled to capacity with 72°F water and placed in an environmental chamber set to -6°F for three (3) hours or until frozen solid. The bladder shall be removed from the environmental chamber and allowed to thaw completely. The system shall be inspected for any damage, degradation, and evidence of leaking due to freezing.

4.5.8 Empty weight (Type I only). The bladder, hose, bite valve assembly and the mask adapter shall be empty of water and weighed with equipment capable of measuring weight to (± 0.01) pounds.

4.5.9 Functional integration with the MOLLE HSC and the AHC (Type I only). An empty dried bladder shall be placed inside of a MOLLE HSC. The bladder inside the MOLLE HSC shall be placed on the scale and the weight shall be tared. The bladder shall be filled to capacity with water then weighed. A fixture may be used to assist in filling the bladder. The minimum net weight shall be 5.8 pounds. Net weight is defined as the filled gross weight of the MPHS (bladder and carrier) minus the tare dry weight of the MPHS. The procedure shall be repeated with the AHC. (Note that a standard MOLLE HSC and the AHC shall be furnished by the contracting officer) (see 6.3).

4.5.10 Leakage (Type I only). A bladder filled to capacity with water shall be laid flat and compressed at the midline with a 20 square inch round anvil on an Instron (or similar) test machine with a 1,000-pound cell. It shall be set at 500-pound cyclic compression. The mechanical valves that restrict the flow of water to the bite valve shall be closed during the test. There shall be no damage or leakage after three (3) cycles at a speed of 20-inches per minute.

4.5.11 Capacity (Type I only). A dry bladder shall be placed on a scale and tared. The bladder shall then be filled to capacity with water. A fixture may be used to assist in filling the bladder. The net weight of the filled bladder must be a minimum of 6.37 pounds. Net weight is defined as the filled gross weight of the bladder minus the tare dry weight of the bladder.

4.5.12 Compatibility with chemical protective masks (Type I only). A bladder shall be filled with water to capacity. The MPHS drink tube and mask adapter shall be connected to the drinking tube of the current chemical protective mask. The mouthpiece of the current chemical protective mask shall be connected to a sealed beaker. The beaker shall be connected to a digital pressure gauge connected to a system supplying a vacuum. The vacuum pressure shall be set to 20,684 Pa (3-psi). Once water begins to flow, time for 1-minute and record the volume of water in the beaker. Repeat test three (3) times. All results shall meet a minimum of 200 ml.

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4.5.13 Identification markings and labeling. Identification markings and labeling shall be examined in accordance with Table II.

4.5.14 User and care instructions. User instructions shall be examined in accordance with Table II.

4.5.15 Unit pack barrier bags.

4.5.15.1 Type I. The Type I unit pack barrier bag shall be tested in accordance with Type II of MIL-PRF-32380 and the exceptions as stated in 3.6.

4.5.15.2 Type II. The Type II, unit pack barrier bag shall be examined in accordance with Table II.

4.5.16 In-process validation (IPV) examination of components. Key critical processes which require 100 percent inspection (see Table I) shall be recorded in a manner to allow for Government QAR validation.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

NOTE: Due to the CBRN protection requirement, the MPHS Type I item is designated as a CSI, specific packaging requirements shall be used for Type I as stated in the contract.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The MPHS is intended to be used by Soldiers in a chemical environment to hydrate when wearing the JSGP Mask.

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6.2 Acquisition documents. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2)
c. The specific issue of individual documents referenced (see 2.2)
d. When first article is required (see 3.1.1)
e. Color required (see 3.1.1, 3.4.1.1, 3.4.2.1 and 3.5)
f. Conformance inspection acceptance quality limits (AQLs) (see 4.3)
g. Inspection conditions (see 4.3.2)
h. Production Lot Testing (see 4.4)
i. Packaging (see 5.1)

6.3 First article. When a first article inspection is required (see 3.1.1), it will be inspected and approved under the appropriate provisions, of the contract, this specification and coordinating drawings. The first article should be a preproduction sample representative of a production lot. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Drawings and standard sample. For access to samples and drawings, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Surveillance program. Thirty-five (35) MPHS will be set aside for surveillance. The end items will be packaged and shipped separately to the address below indicating the point contact stated in the contract:

Traffic Manager
JEFS TEST MANAGEMENT
SERVICES 120 MAPLE ST
LONDON, OH 43140
740-852-5256

6.6 Government approved laboratories for the MPHS. Please contact the Program Management Office for a list of approved laboratories. PMO Contact information:

Product Manager Soldier Clothing and Individual Equipment
5981 13th Street, Building 1155
Fort Belvoir, VA, 22060
Load Carriage and Combat Equipment Team

6.7 Superseding documents. This document supersedes GL-PD-15-03 Multi-Purpose Personal Hydration System (all revisions) and will be used in all future procurements.

6.8 Types. Previous types cited in GL-PD-15-03 Multi-Purpose Personal Hydration System have been changed to the following in this specification:

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GL-PD-15-03	MIL-DTL-32723
Type I - CBRN Hydration System, MPHS.	Type I - CBRN Hydration System
Type II - CBRN Hydration Training Kit, MPHS.	Type II - CBRN Hydration Training Kit
Type IIA - Bite valve with cover.	Type IIA - Bite valve with cover
Type IIB - DELETED	Type IIB - Mask adapter
Type IIC - M50 Series Mask adapter.	Type IIC - Drink tube adapter
Type IID - Drink tube adapter.	

6.9 Cross-reference. The following Purchase Descriptions are being referenced in this document.

CO-PD-02-02 - Modular Lightweight Load-Carrying Equipment (MOLLE)
GL-PD-19-04 - Aviation Hydration Carrier (AHC)

(Copies of these documents are available through the contracting activity.)

6.10 Subject term (key word) listing.

Bladder
Chemical, Biological, Radiological, Nuclear (CBRN) Drink tube
Mask Adapter
Modular Lightweight Load-Carrying Equipment (MOLLE) Training Kit

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APPENDIX A

Multi-purpose Personal Hydration System (MPHS)
System Level - Chemical Warfare Agent Test Method

A.1 SCOPE

A.1.1 Scope. This appendix covers the testing protocol to determine the protective capability of the MPHS to chemical warfare agents. This appendix is mandatory part of this specification.

A.2 TEST PROTOCOL

A.2.1 Test objectives. The entire hydration system shall be evaluated to determine the Hydration System Chemical Warfare Agent (CWA) protective capability after six hours of exposure. Specifically, the calculated concentration of any CWA with its respective hydrolysis products found to permeate/penetrate the test samples shall be below that which is allowed by Military Exposure Guidelines (MEGs) for 15 liters of water consumption by a 70-kg human per for up to seven (7) days. Each test shall be conducted for six (6) hours, as these systems are designed to provide chemical protection for that amount of time.

The following material conditions shall be evaluated against the criteria in Table A-I.

TABLE A-I. Test Criteria.

Item	Test Criteria		
1	Material shall be received in an acceptable condition for System testing.		
2	Total calculated concentration of CWA and associated hydrolysis product to permeate/penetrate shall be compared to the Military Exposure Guidelines for drinking water standards after a 6-hour test: MEGs for MPHS CWAs		
	CWA	5 L/day MEG (mg/L)	15 L/day MEG (mg/L)
	GD	0.012	0.004
	HD	0.14	0.047

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The criteria against which the items shall be evaluated are based on Military Exposure Guidelines from:

1. U.S. Army Public Health Command, Technical Guide 230, Environmental Health Risk Assessment and Chemical Exposure Guidelines for Deployed Military Personnel

2. Department of the Army, Technical Bulletin Medical 577, Sanitary Control and Surveillance of Field Water Supplies.

A.2.2.2 Test equipment.

a. The test fixture shall be comprised of components that work in unison to challenge, control and sample the test items. These components are the test chamber, agent application device and sample collection manifold.

i. Test Chamber. The test chamber shall be a stainless steel sufficiently large to accommodate eight (8) samples, in one (1) trial. The test chamber shall be capable of meeting and recording the required environmental conditions at least every five (5) minutes.

ii. Agent application. The purpose of the dispenser is to reproducibly and safely apply chemical warfare agent to the test items. Agent shall be applied using a syringe capable of dispensing 10 (±1) µL drops.

iii. Vapor sample collection manifold. A sample collection manifold shall be used to collect agent vapor concent0rations from the chamber environment at time zero, 1-hr, 2-hr, 3-hr, 4-hr, 5-hr, and 6-hr intervals.

b. Analytical Instrumentation.

i. Gas Chromatography/Mass Spectrometry (GC/MS). A GC/MS system shall be used to quantify any HD, or TGD present in the organic extract of the water samples taken from each test item. All GCs shall be calibrated for the specified CWA prior to any sample analysis.

ii. Liquid Chromatography/Mass Spectrometry. A LC-MS system shall be used for the analysis of the aqueous phase of the extraction for hydrolysis product of HD only. The primary hydrolysis product for HD is thiodiglycol (TDG).

A.2.2.3 Test procedure.

a. Test Item Receipt Inspection

Upon receipt, each test item shall be inspected and verified to be undamaged and in working condition. A visual inspection and leak test shall be made of the test item prior to testing to verify the physical integrity is intact. The leak test includes filling the bladders with water

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and visually looking for any water leaks. Test items that have rips, tears or other physical defects shall not be tested.

b. Environmental Conditioning

The test items shall be filled with one (1) liter of American Chemical Society (A.C.S.) reagent grade water (CAS 7732-18-5). One (1) liter of water is used in order to meet the MEG detection limits. All air shall be removed from the bladder and hose of each test item. The test items shall be placed in the test chamber and test chamber shall be sealed and the environmental conditions will be brought to test conditions (Table A-II).

TABLE A-II. MPHS Full System Test Conditions.

Temperature (Chamber)	32.2 (±1.7)°C / 90.0 (±3.0)°F)
Relative Humidity	80 (±8)%
Chamber Pressure Differential	Between 0.0 and -1.0 iwg
Chamber Airflow	50 (±5) liters per minute

c. Agent Contamination

The CWA challenge to each system shall be based on a $10g/m^2$ liquid challenge criterion. The test items shall be placed on shelves located inside the test chamber, oriented horizontally for the testing process. The top surface of the carrier and the exposed section of the hose shall be contaminated with CWA. The CWA challenge shall be calculated based upon the surface area of the MPHS bladder and drink hose. The CWA challenge by weight shall be roughly 1.23g for the MPHS. Contamination by volume for the MPHS shall be 970µL of HD and 1200µL of TGD. All CWA shall be applied as individual 10-µL droplets. The MPHS shall have 24 drops applied to the hose and 73 to the carrier for HD,
30 drops applied to the hose and 90 to the carrier for TGD.

The locations shown in Figures A-1 through A-2 shall be used as target challenge areas when applying the contamination to the MOLLE carrier and hoses.

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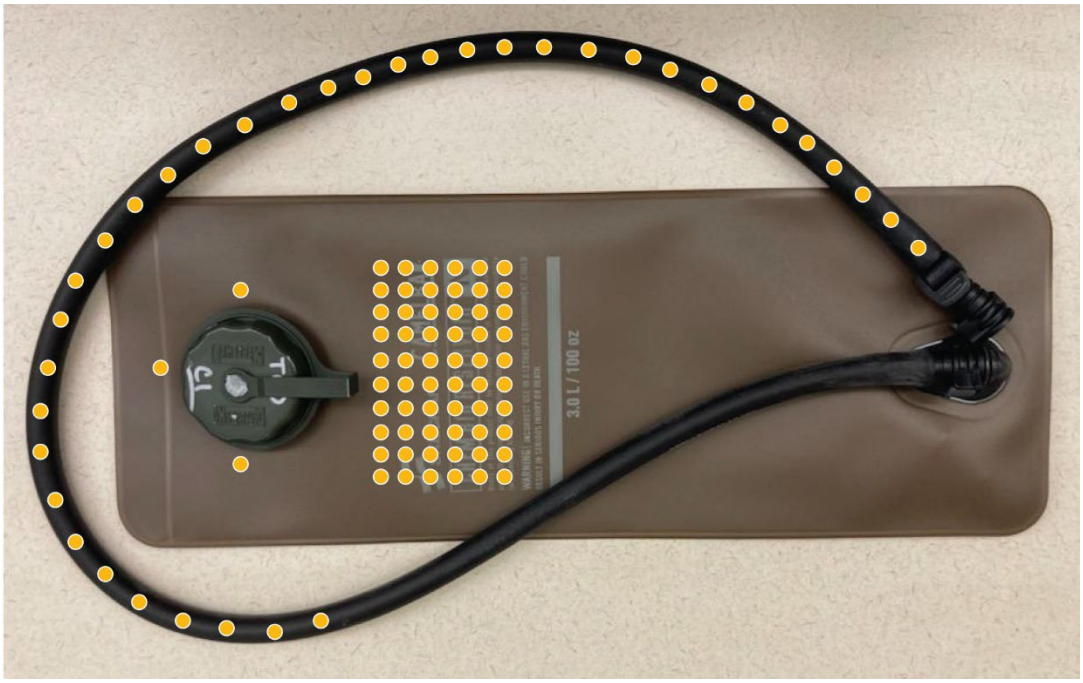


FIGURE A-1. HD Pattern.

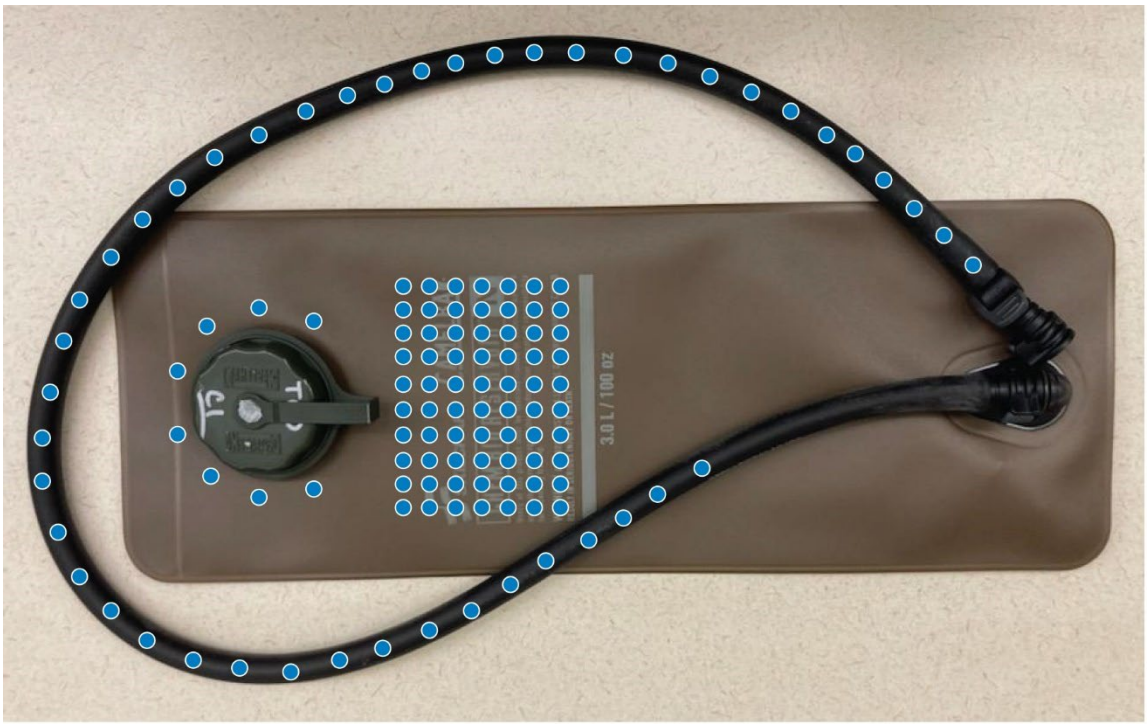


FIGURE A-2. TGD Pattern.

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Each agent lot shall undergo a purity analysis prior to use. This test program shall only use CWA with a purity of greater than 85 percent. A vapor challenge of 10 mg/m^2 or higher shall be created in the chamber as a result of evaporation throughout the duration of each test. The sample collection manifold shall be used to measure agent chamber concentration at a minimum of one (1) per hour.

During the agent operation, the start time of the first test item contaminated shall be recorded as the spiking start time. Once the first test item shall be contaminated, the time shall be recorded as the end of contamination. This process shall continue for all test items.

After contamination, the vacuum pump and the sampling manifold control system shall be turned on to initiate the start of test. The test start time shall be annotated. During the test, all flow, sampling times, temperature, relative humidity, and pressure shall be electronically recorded by computer software. The data interval shall be no greater than five (5) minutes.

Contamination controls shall be completed either prior to or just after the start of the trial.

d. Sampling, Extraction and Analysis

There shall be two (2) samples taken at the six (6) hour point for each tested item, one (1) representing the drinking hose and the other representing the bag. Samples shall be collected utilizing the access ports connected to the test items hoses.

The sample from the drink tube or hose shall be taken first. This shall be done by opening the test item valve and removing 35 mL of water using a vacuum of no more than three (3) pounds per square inch. After the hose sample has been removed, the remaining water in the bladder shall be removed using a vacuum of no more than three (3) pounds per square inch.

There shall also be sorbent tube analysis of chamber concentration per trial. Solid sorbent tubes (SSTs) packed with Chromosorb 106 shall be used to collect TGD and HD.

e. Trial Completion

Trials shall be terminated after six (6) hours and the test items sampled as described above.

After sampling has been completed, the test fixture shall be air washed to lower the agent concentration within the fixture. A hazard analysis may be needed by the test organization based upon the volume of agent to be used per trial. Based on the results of the hazard analysis, a determination shall be made regarding the amount of time required for the air wash of the chamber to clear down the vapor concentration prior to the removal of the test items. After the required minimum time by the test organization, the MPHS test items shall be removed from the test chamber and placed in the appropriate decontamination solution and disposed of as hazardous waste.

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A.2.2.4 Quality Control.

a. Control Items

Each trial shall have one positive control and one negative control. The purpose of the positive control shall be to identify any gross process drift in testing and analytical procedures. The positive control shall be an IV bag filled with 1000 mL of sterile water. A sample line shall be connected to the positive control to remove water at the conclusion of the six (6) hour trial. The positive controls shall be contaminated on the exterior of the bag at the same 10 g/m^2 as the test items. Ten-µL drops shall be evenly deposited over the surface for both HD and TGD. HD shall require 25 droplets and TGD 32 droplets. Water samples from the positive control shall be collected six (6) hours after agent contamination, extracted, and analyzed the same as the samples taken from the test items.

The purpose of the negative control shall be to identify any agent carryover caused by positive or negative analytical interference. The negative control shall be an IV bag or MPHS filled with 1000 mL of sterile water. A sample line shall be connected to the negative control to remove water at the end of the six (6) hour test. The negative control shall be located outside of the chamber and shall not be contaminated with CWA. Water from the negative control shall be extracted and analyzed identically to the bladder samples.

b. Contamination Controls

The purpose of the contamination control is to confirm the quantity of agent applied per droplet. During the contamination of the MPHS test items for a trial, three control samples shall be generated. Two (2) 10-µL drops of CWA shall be applied to the coupons and the coupons shall be placed into jars filled with 100 mL chloroform. The jars shall either be shaken by hand or on a shaker table, then transferred to GC vials for analysis. Results shall be used to calculate the actual contamination density with which the MPHS test items were challenged.

c. Instrument Calibration

Instruments used on this program shall be calibrated prior to test initiation. The instruments shall be checked to ensure the calibration is maintained at the conclusion of the program or after the time period specified by the quality policies of the laboratory, whichever comes first. Procedures to calibrate and check calibration shall be in accordance with the manufacturer's specifications.

d. Data Management and Quality Assurance

The objective of data management and quality assurance (QA) procedures is to ensure the completeness and validity of the data acquired and processed throughout the program. All data generated during the program shall be accounted for in controlled laboratory records.

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The laboratory records shall be used to document all aspects of a test (such as visual observations) that are not being automatically recorded by computerized control system, and to thoroughly document test trials, chain of custody forms, and samples that were run within those trials.

Data QA shall be achieved by implementing the following procedures:

- Daily verification of procedures and instrument results shall be entered into the laboratory records (signature by Tester and Supervisor shall be required).
- Daily verification that all data generated during a test trial is accounted for and traceable to the laboratory records.
- Daily verification that all test controls are functioning properly.
- Analytical Standards verification.
- Electronic Control and Password protection of all data.

A.3 RESULTS AND ANALYSIS

A.3.1 Measurements.

The following measurements shall be made during each trial:

- Agent mass collected in each sample
- Pressure differential between the inside and outside of the chamber
- Chamber air temperature and relative humidity
- Contamination controls
- Time of sample collection
- Flow rate into chamber
- Agent concentration inside the chamber throughout test

A.3.2 Results. The results for the MPHS test items shall be reported as mass of agent per liter. Calculate the pass/fail MEG based upon both the bladder and hose samples. Table A-III shows an example of the calculations.

TABLE A-III. Total detectable agent calculations.

Test agent	Analytical chemical	Water location	Water vol. (mL)	Sub sample (mL)	Extract (mL)	LOD analytical result (ng/mL)	Total mass (ng)	Full PHS (L)	Full PHS (L)	Total detectable test agent (µg/mL)
HD 1/	HD	Bladder	965	10	2	2.0	1930	2.896	0.69	2.49
		Hose	35	10	2	2.0	70			
	TDG	Bladder	965	1		4.0	5025	2.896	0.69	
		Hose	35	1		4.0	182			
TGD 2/	GD	Bladder	965	10	2	2.4	2316	2.896	0.83	0.83
		Hose	35	10	2	2.4	84			

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1/ HD MW (g/mL) = 159.07
2/ TDG MW (g/mL) = 112.18

Total Mass

Equation 1. Agent calculations Bladder and Hose: Water Volume X

Analytical Result = ng

Equation 2. Hydrolysis Product (HP) Bladder and Hose:

$$(\text{Water Volume X Analytical Result}) \text{ X } \frac{\text{Agent MW}}{\text{HPMW}} = \text{ng}$$

PHS Concentration

Equation 3. Concentration Determined per Analyte:

$$\frac{\text{Hose Mass + Bladder Mass}}{\mu\text{g/L (FullPHS:2.896L) X 1000}} =$$

Total Detectable Agent

Equation 4. Total Detectable Test Agent

PHS Concentration agent + PHS Concentration HP = µg/L

a. Detection Limits and Surrogate Values for Non-detected Chemicals. The Limit of Quantitation (LOQ) shall be set at or above the concentration of the lowest initial calibration standard (USAPHC, TG 230). The required Limits of Quantitation (LOQ) for TGD, HD, and TDG are shown below in Table A-IV.

TABLE A-IV. Minimum limits of quantitation.

Analysis	Analyte	LOQ (ng/mL)	LOD (ng/mL)	MDL (ng/mL)
GC- MS/MS	HD	4.0	2.0	0.63
GC-MS/MS	TGD	4.0	2.4	1.1
LC-MS/MS	TDG	22	4.0	0.94

b. Data that falls below the LOQ shall be reported as described in Table A-V using Method A. The definitions and methods for determining the Detection Limit (DL), Reporting Limit (RL), and Limit of Detection (LOD), are found in paragraph 3.4.4.3 of

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Technical Guide 230 from US Army Public Health Command, Revision 2013. The test lab shall report DL, RL, LOD and LOQ along with the methods for determining each.

TABLE A-V. Relationship between detection limits, data flags, and reported results.

Possible Laboratory Outcomes		Laboratory Reported Results		
		Reported Data Flag	Reported Value	
Result	The concentration is…		Method A	Method B
Not detected	Less than the DL	U	LOD	LOQ
Detected	Between the DL and the RL	J	Measured value	Measured value
Detected	Equal to or greater than the RL	No flag	Measured value	Measured value

Reference: USAPHC Technical Guide 230, Revision 2013.

Custodian:
Army - GL

Preparing activity:
DLA-CT

(Project: 8465-2022-005)

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